**20** What is the electrostatic force between a thorium nucleus of charge $1.4 \times 10^{-17}$ C and an alpha particle when they are separated by a distance of 0.10 pm?

**A** $4.0 \times 10^{-20}$ N **B** $4.0 \times 10^{-13}$ N **C** 4.0 N **D** 8.0 N

Ans: **C**

Potential is a scalar.

$$F = \frac{Q_{Th}Q_{\alpha}}{4\pi\varepsilon_0 r^2} = \frac{(1.4\times10^{-17})(2\times1.6\times10^{-19})}{4\pi(8.85\times10^{-12})(0.10\times10^{-12})^2} = 4.0 \text{ N}$$

**A:** forgot to convert pm

**B:** forgot to divide by $r^2$

**D:** took helium charge as 4e.

**21** A conducting liquid fills a cylindrical metal case to a depth *x* as shown below. The case has an